

- Description

This module manages a smart weather station's rainfall intensity monitoring system, ensuring that:

1. The system continuously monitors the current rainfall intensity (in mm/h).
2. If the rainfall intensity exceeds a predefined threshold, a heavy rain alert is triggered.
3. If the rainfall intensity is below the threshold, the system indicates normal rainfall conditions.

This system helps provide timely alerts for heavy rain, enabling proactive weather response measures.

- Specification

module SmartRainfallMonitoring

var currentRainfall, rainfallThreshold: real

process updateRainfall(newRainfall: real) returns (currentRainfall: real)

functional scenario form:

currentRainfall > rainfallThreshold && output "ALERT: Heavy rain detected (" + currentRainfall + " mm/h). Please take precautionary measures.", Heavy rain alert is triggered.

||

currentRainfall <= rainfallThreshold && output "Rainfall intensity is normal (" + currentRainfall + " mm/h).", Rainfall intensity is normal.

end process

end module

- Java Code

```
package code;
public class c103_SmartRainfallMonitor {
    private double rainfallThreshold;
    private double currentRainfall;
    public c103_SmartRainfallMonitor(double rainfallThreshold, double initialRainfall) {
        this.rainfallThreshold = rainfallThreshold;
        this.currentRainfall = initialRainfall;
    }
    public void updateRainfall(double newRainfall) {
        currentRainfall = newRainfall;
        System.out.printf("Current Rainfall Intensity: %.2f mm/h\n", currentRainfall);
        if (currentRainfall > rainfallThreshold) {
            System.out.println("ALERT: Heavy rain detected! Please take precautionary measures.");
        } else {
```

```

        System.out.println("Rainfall intensity is normal.");
    }
}

public double getCurrentRainfall() {
    return currentRainfall;
}

public static void main(String[] args) {
    c103_SmartRainfallMonitor monitor = new c103_SmartRainfallMonitor(
20.0, 10.0);

    // testcase-VT:
//    monitor.updateRainfall(15.0);
//    monitor.updateRainfall(25.0);
//    monitor.updateRainfall(18.0);

    // testcase-FT:
//    monitor.updateRainfall(2.743040779495874E303);
//    monitor.updateRainfall(1.071500304489605E301);
//    monitor.updateRainfall(-1.7976931348623157E308);

    // testcase-Z3:
//    monitor.updateRainfall(15.0);
//    monitor.updateRainfall(25.0);
//    monitor.updateRainfall(18.0);

    // testcase-UVT:
    monitor.updateRainfall(15.0);
    monitor.updateRainfall(25.0);
    monitor.updateRainfall(18.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c103_SmartRainfallMonitor.java	100.0 %	2	0	2
▼ c103_SmartRainfallMonitor	100.0 %	2	0	2
● updateRainfall(double)	100.0 %	2	0	2
● c103_SmartRainfallMonitor(doubl		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,830	21,884
▼ src	0.2 %	54	21,830	21,884
▼ code	0.2 %	54	21,830	21,884
▼ c103_SmartRainfallMonitor.java	94.7 %	54	3	57
▼ c103_SmartRainfallMonitor	94.7 %	54	3	57
● updateRainfall(double)	100.0 %	29	0	29
● c103_SmartRainfallMonitor(doul	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

4. Fuzz testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c103_SmartRainfallMonitor.java	100.0 %	2	0	2
▼ c103_SmartRainfallMonitor	100.0 %	2	0	2
● updateRainfall(double)	100.0 %	2	0	2
● c103_SmartRainfallMonitor(doul		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,830	21,884
▼ src	0.2 %	54	21,830	21,884
▼ code	0.2 %	54	21,830	21,884
▼ c103_SmartRainfallMonitor.java	94.7 %	54	3	57
▼ c103_SmartRainfallMonitor	94.7 %	54	3	57
● updateRainfall(double)	100.0 %	29	0	29
● c103_SmartRainfallMonitor(doul	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c103_SmartRainfallMonitor.java	100.0 %	2	0	2
▼ c103_SmartRainfallMonitor	100.0 %	2	0	2
● updateRainfall(double)	100.0 %	2	0	2
● c103_SmartRainfallMonitor(doul		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	54	21,830	21,884
▼ src	0.2 %	54	21,830	21,884
▼ code	0.2 %	54	21,830	21,884
▼ c103_SmartRainfallMonitor.java	94.7 %	54	3	57
▼ c103_SmartRainfallMonitor	94.7 %	54	3	57
● updateRainfall(double)	100.0 %	29	0	29
● c103_SmartRainfallMonitor(doul	100.0 %	9	0	9
● main(String[])	100.0 %	16	0	16

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c103_SmartRainfallMonitor.java	100.0 %	2	0	2
▼ c103_SmartRainfallMonitor	100.0 %	2	0	2
● updateRainfall(double)	100.0 %	2	0	2
● c103_SmartRainfallMonitor(doul		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 94.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool	 0.2 %	54	21,830	21,884
▼  src	 0.2 %	54	21,830	21,884
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▼  c103_SmartRainfallMonitor.java	 94.7 %	54	3	57
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